

Thalassemia major

T. Sujitha

Name: Tisha

Age: 11 years

Sex: Female

Address: Mandaveli (Nativity: Rajasthan)

Informant: Mother

Reliability: Good

Birth order: second, born by non-consanguineous marriage

Chief complaints:

A known case of Thalassemia major diagnosed at 6th month came for blood transfusion.

H/O presenting illness:

- She was apparently normal before 5 months of age.

- At 5th month, she developed pallor, which was not treated due to mother's negligence.

- At 6th month, she developed an episode of fever, cold and wheeze. During investigation, the Hb was found to be 4.5 gm/dl and was diagnosed as Thalassemia major.

- She is on blood transfusion at each month after the diagnosis.
 - Inter-transfusion period is uneventful
 - Total transfusion volume — (quantity)
 - Blood transfused initially, once in — days and now once in 30 days. (frequency of blood transfusion)
 - No H/O lethargy
 - No H/O sleepiness
 - No H/O inability to concentrate
 - No H/O poor school performance
 - No H/O stunting
 - No H/O polyuria
 - No H/O polyphagia
 - No H/O polydipsia
 - No H/O convulsions
 - No H/O breathlessness
 - No H/O palpitations
 - No H/O pedal edema
- } 2/o
 } anaemia
 } (due to blood transfused anaemia)
- } 2/o
 } pituitary
- } - pancreas
 } (diabetes insipidus)
 } parathyroid
- } due to iron overload
 } (transfusion related complication)
- } CCF
 } due to anaemia

Natal history:

- Full term, normal vaginal delivery
- Birth weight: 2.9 kgs
- Cried immediately after birth
- Breastfed within $\frac{1}{2}$ hr

Neonatal history:

- Exclusively breastfed for 6 months
- No H/O neonatal jaundice
- No H/O respiratory distress syndrome

Developmental milestones:

- No developmental delay at this age / Development milestones attained for appropriate age

But initially there was delay in development in gross motor i.e. Head control attained at 7 months
(N - 3-4 months)

ii) Stood with support at 1 yr
(N - 9 months)

iii) Stood without support at $1\frac{1}{2}$ yrs
(N - 1 yr)

Developmental milestones for 11 yrs:

- Fine motor : Can copy cube
- Speech & Language : Reads a story
Uses telephone
- Social & adaptive skills : Assumes responsibility for household care

Diet history: By 24 hr recall method

		Calorie (kcal)	Protein (g)
Morning	Bread - 4	$4 \times 85 = 340$	8
8:30 am	Tea	75	0.9
Afternoon	Rice - 1 cup	170	7
12:30 pm	Sambhar - $\frac{1}{2}$ cup	55	3
Evening	Milk - 1 cup	180	3.2
5:30 pm	Biscuit - 4	$4 \times 20 = 80$	1
Night	Chappathi - 2	$2 \times 60 = 120$	6
9:30 pm	Dal gravy - 1 cup	350	4.7
	Total	<u>1370</u>	<u>33.8</u>

Calorie gap : 640 kcal

Protein gap : 6.2 g

(Exp: 2010 kcal
40 g)

Calorie : Total against expected with 30% calorie deficit
(1270 against 2010 kcal)

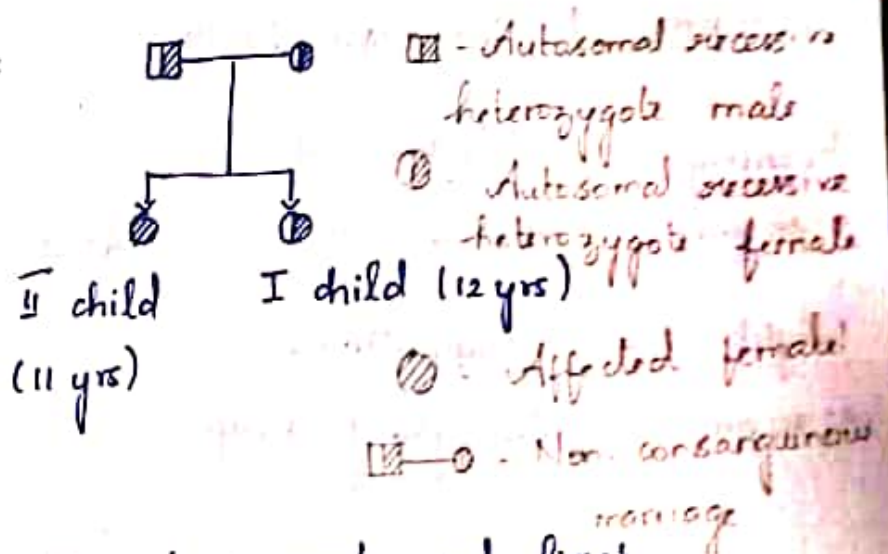
Protein : Total against expected with 15% protein deficit
(33.8g against 40g)

Immunization history:

- Immunized up to date
- Lastly immunized with TT at 10 years
- Pneumococcal vaccine given after splenectomy
(Meningococcal vaccine - given/not)

Family history:

Pedigree chart:



- H/o Thalassemia minor in parents and first child which was diagnosed retrospectively after diagnosing Thalassemia major in second child.

Contact history:

No H/O contact with known case of TB

Socio-economic history:

Housing: Pucca house

Total no. of family members: 4

No. of rooms: 3

Adequately ventilated & lighted

No overcrowding

Separate kitchen

Can water for drinking

Proper sewage disposal

Proper sanitation

SEB:	Education	: High School	- 4
(Head of the family)	Occupation	: Shop owner	- 5
	Family income	: 16021 - 21360	- 6
			<u>15</u>

According to Modified Kuppusamy scale, the family belongs to lower middle (III) socio-economic class.

Summary:

A 11 yr old female child Isha from Mandaveli, a known case of Thalassemia major with H/o lethargy, with nutritional status of caloric deficit, of protein deficit, a previous H/o splenectomy surgery and immunized upto date has come for blood transfusion.

General examination:

The Child is conscious, oriented, active, moderately nourished.

- Afebrile
- Pallor present
- No icterus
- No cyanosis
- No clubbing
- No lymphadenopathy
- No generalised / pedal edema

Vitals: (N - for 11 yrs)

Pulse rate : 75/min (N - 60-100/min)

Rhythm : regular

Volume : normal

Character : normal

No radio-femoral or radio-radial delay

Respiratory rate : 18/min (N - 12 to 18/min)

BP : 100/70 mmHg (N - 90-110 / 60-90)

Anthropometry:

Ht : 130 cm

Wt : 29 kgs Exp : 35 kgs } for 11 yrs (girls)

BMI : 17 (Classified as underweight)

Head to foot examination:

- Face :
 - Frontal bossing present
 - Widely spaced eyes
 - Flat nasal bridge
 - Upper teeth projected forwards
 - Prominent malar bones
- (Hemolytic facies)

- Neck : normal
- Chest : normal
- Abdomen : Umbilicus in midline
Splenectomy scar seen in the left hypochondrium
- Limbs : normal

Systemic examination:

Abdomen:

Inspection : - Abdomen not distended, all quadrants moves equally on respiration

- Umbilicus in midline
- No dilated veins, visible pulsations
- Flanks free
- Hernial orifices free
- Splenectomy scar seen extending from midline to left hypochondrium about 10 cm in length.

Palpation: - Liver is palpable in right hypochondrium in mid clavicular line, 4 cm below the subcostal margin

- Surface - smooth
- Firm in consistency
- Edges - rounded & well defined

Percussion: On percussion, the liver span was found to be 4 cm in mid clavicular line.
(11-12 cm) for 4 cm

Auscultation: Normal bowel sounds heard

Cardiovascular system:

- Apical impulse (In Cardiomegaly due to iron deposition it is downward & outward)

In Anaemia - hyperdynamic

- first, second heart sounds heard
- No added sounds
- No murmurs

Respiratory system:

- Normal vesicular breath sounds heard
- No added sound

Central Nervous system:

- No focal neurological deficit

Diagnosis:

A 11 yrs old female child Isha, of mixed birth order, born out of non-consanguineous marriage, from Mandaveli is a known case of Thalassemia major, with hemolytic features, hepatomegaly and pallor, splenectomy done previously, with nutritional status of caloric deficit, of protein deficit and BMI-17, Immunized upto date has come for blood transfusion.

Investigations:

- CBC
- Peripheral smear
- Serum ferritin
- Liver function tests (1 bilirubin. enzymes normal but)
- Reticulocyte count
- X-ray
- Blood urea, creatinine
- Hb electrophoresis

Management:

- Blood transfusion
- Iron chelating agents
- Hematopoietic stem cell transfusion
- Genetic counselling
- Nutritional advice (↓ Iron intake spinach
↑ Tea - it decreases iron absorption)

leucocyte depleted cells -